

Algorithm Analysis & Complexity Theory

Sample Exam: Recurrence Relations

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Difficulty Levels: Foundation, Conceptual, Application, Advanced

Question 1 [Difficulty: Conceptual]

Solve the recurrence $T(n) = 2T(n/2) + n$ using the recursion tree method.

Solution:

Recursion Tree Method:

Level 0: n Cost: n

Level 1: $2 \cdot (n/2)$ Cost: n

Level 2: $4 \cdot (n/4)$ Cost: n

Level 3: $8 \cdot (n/8)$ Cost: n

...

Level i : $2^i \cdot (n/2^i)$ Cost: n

Tree height: $\log_2(n)$ (when we reach $T(1)$)

Total cost = sum of costs at each level

$= n + n + n + \dots + n$ ($\log n$ times)

$= n \cdot \log n$

Therefore, $T(n) = \Theta(n \log n)$ ■

Question 2 [Difficulty: Application]

Use the Master Theorem to solve $T(n) = 4T(n/2) + n^2$.

Solution:

Master Theorem for $T(n) = aT(n/b) + f(n)$:

Given: $T(n) = 4T(n/2) + n^2$

- $a = 4$
- $b = 2$
- $f(n) = n^2$

Calculate: $\log_b(a) = \log_2(4) = 2$

Compare $f(n) = n^2$ with $n^{\log_b(a)} = n^2$:

Case 2 applies: $f(n) = \Theta(n^{\log_b(a)} \cdot \log^k(n))$ where $k = 0$

Therefore: $T(n) = \Theta(n^2 \log n)$ ■